DBMS MINIPROJECT

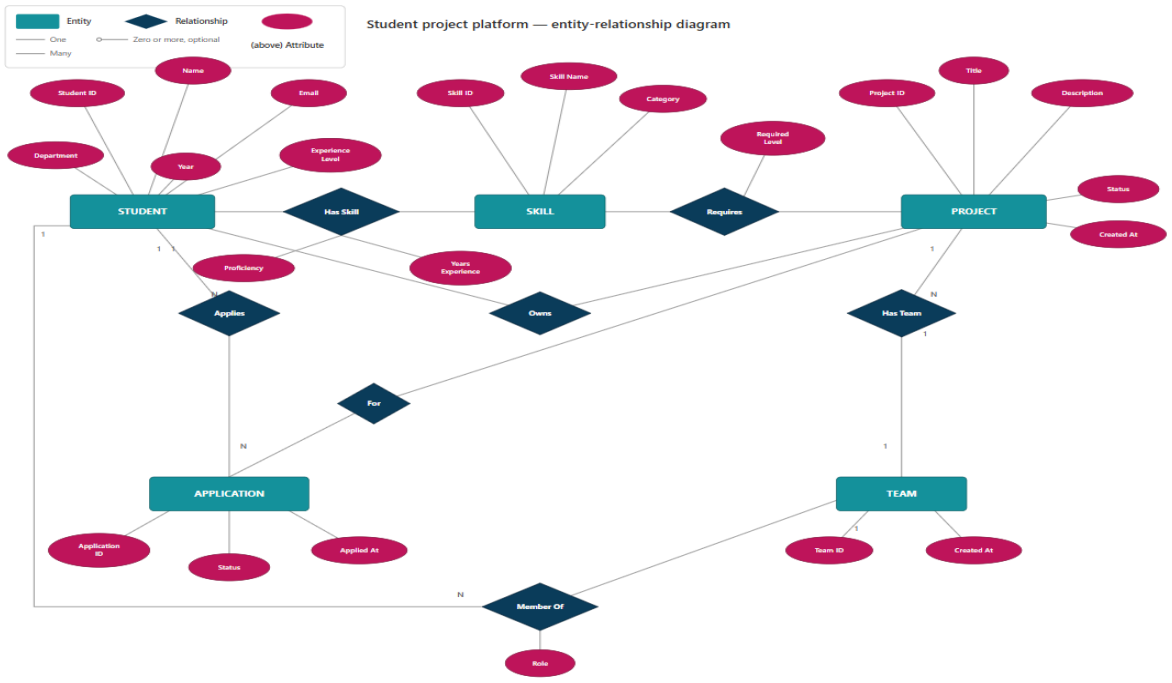
Team Members:

1. Name - Shamanth Shekar K SRN - PES1UG24AM258
2. Name - Satyanarayana K SRN - PES1UG24AM257

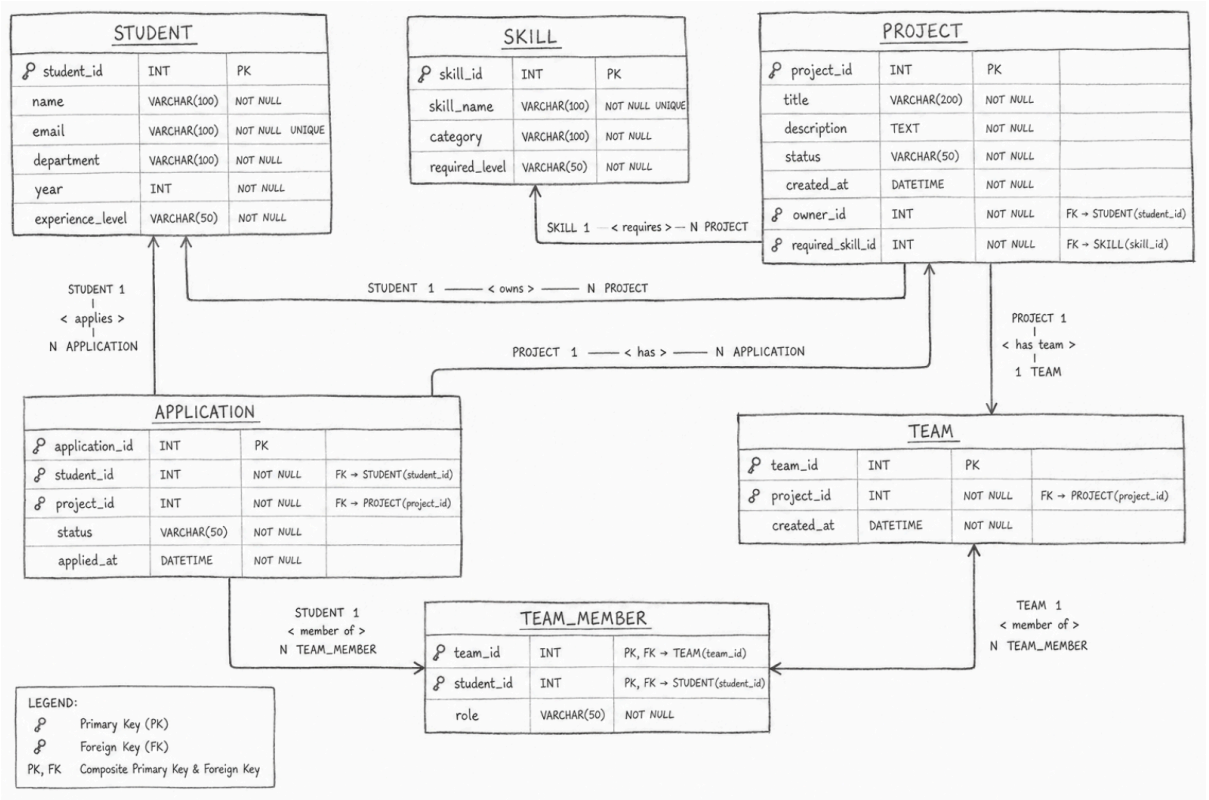
Topic - SkillSync

Description - SkillSync is a web-based platform that connects students based on their skills, interests, project requirements, availability, and experience. Students can create skill profiles, publish projects, specify the skills they need, and discover suitable teammates or collaborators.
The interesting part is that SkillSync goes beyond simple filtering. The system combines structured SQL-based matching—such as skill level, availability, department, experience, and project requirements—with semantic similarity search to understand descriptions such as “I need someone who knows machine learning and can work on computer vision” and find students whose profiles are conceptually relevant even when they don't use exactly the same keywords.
The system will maintain student profiles, skills, projects, applications, teams, availability, ratings, and activity history in PostgreSQL, while pgvector can be used for semantic representations of skill/project descriptions. The application will provide separate interfaces for Students and Administrators, including dashboards, project discovery, intelligent matching, applications, team formation, search/filtering, and analytics.

ER-Diagram:



Relational Diagram:



ScreenShots:

```
mysql> SELECT DATABASE();
+---------------+
| DATABASE()    |
+---------------+
| skillsync_db  |
+---------------+
1 row in set (0.000 sec)
```

```
mysql> DESCRIBE STUDENT;
+------------------+--------------+------+-----+---------+----------------+
| Field            | Type         | Null | Key | Default | Extra          |
+------------------+--------------+------+-----+---------+----------------+
| student_id       | int          | NO   | PRI | NULL    | auto_increment |
| name             | varchar(100) | NO   |     | NULL    |                |
| email            | varchar(150) | NO   | UNI | NULL    |                |
| department       | varchar(100) | NO   |     | NULL    |                |
| year             | int          | NO   |     | NULL    |                |
| experience_level | varchar(30)  | NO   |     | NULL    |                |
+------------------+--------------+------+-----+---------+----------------+
6 rows in set (0.002 sec)
```

```
mysql> DESCRIBE SKILL;
+------------+--------------+------+-----+---------+----------------+
| Field      | Type         | Null | Key | Default | Extra          |
+------------+--------------+------+-----+---------+----------------+
| skill_id   | int          | NO   | PRI | NULL    | auto_increment |
| skill_name | varchar(100) | NO   | UNI | NULL    |                |
| category   | varchar(100) | NO   |     | NULL    |                |
+------------+--------------+------+-----+---------+----------------+
3 rows in set (0.003 sec)
```

```
mysql> DESCRIBE STUDENT_SKILL;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| student_id       | int          | NO   | PRI | NULL    |       |
| skill_id         | int          | NO   | PRI | NULL    |       |
| proficiency      | varchar(30)  | NO   |     | NULL    |       |
| years_experience | decimal(4,1) | NO   |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
4 rows in set (0.003 sec)
```

```
mysql> DESCRIBE PROJECT;
+-------------+-------------+------+-----+-------------------+-------------------+
| Field       | Type        | Null | Key | Default           | Extra             |
+-------------+-------------+------+-----+-------------------+-------------------+
| project_id  | int         | NO   | PRI | NULL              | auto_increment    |
| title       | varchar(150)| NO   |     | NULL              |                   |
| description | text        | YES  |     | NULL              |                   |
| owner_id    | int         | NO   | MUL | NULL              |                   |
| status      | varchar(30) | NO   |     | NULL              |                   |
| created_at  | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-------------+-------------+------+-----+-------------------+-------------------+
6 rows in set (0.004 sec)
```

```
mysql> DESCRIBE PROJECT_SKILL;
+-----------------+-------------+------+-----+---------+-------+
| Field           | Type        | Null | Key | Default | Extra |
+-----------------+-------------+------+-----+---------+-------+
| project_id      | int         | NO   | PRI | NULL    |       |
| skill_id        | int         | NO   | PRI | NULL    |       |
| required_level  | varchar(30) | NO   |     | NULL    |       |
+-----------------+-------------+------+-----+---------+-------+
3 rows in set (0.003 sec)
```

```
mysql> DESCRIBE APPLICATION;
+----------------+-------------+------+-----+-------------------+-------------------+
| Field          | Type        | Null | Key | Default           | Extra             |
+----------------+-------------+------+-----+-------------------+-------------------+
| application_id | int         | NO   | PRI | NULL              | auto_increment    |
| project_id     | int         | NO   | MUL | NULL              |                   |
| student_id     | int         | NO   | MUL | NULL              |                   |
| status         | varchar(30) | NO   |     | NULL              |                   |
| applied_at     | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+----------------+-------------+------+-----+-------------------+-------------------+
5 rows in set (0.003 sec)
```

```
mysql> DESCRIBE TEAM;
+------------+-----------+------+-----+-------------------+-------------------+
| Field      | Type      | Null | Key | Default           | Extra             |
+------------+-----------+------+-----+-------------------+-------------------+
| team_id    | int       | NO   | PRI | NULL              | auto_increment    |
| project_id | int       | NO   | MUL | NULL              |                   |
| created_at | timestamp | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------+-----------+------+-----+-------------------+-------------------+
3 rows in set (0.004 sec)
```

```
mysql> DESCRIBE TEAM_MEMBER;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| team_id    | int         | NO   | PRI | NULL    |       |
| student_id | int         | NO   | PRI | NULL    |       |
| role       | varchar(50) | NO   |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
3 rows in set (0.003 sec)
```